1

FISH BIODIVERSITY AND METAL

INTRODUCTION

Freshwater ecosystems are vital for sustaining biodiversity and supporting human activity and recreation. **Bioaccumulation is the gradual accumulation of substances faster than it can be expelled**, such as pesticides or other chemicals, in an organism

Does the presence of heavy metals in bodies of freshwater correlate with decreased fish biodiversity?

CONTEXT

“Heavy metals, the **most potent contaminants** of the environment” **[8]**

Toxic heavy metals enter the food chain via industry and bioaccumulate in fish, raising major health concerns. **[8]**

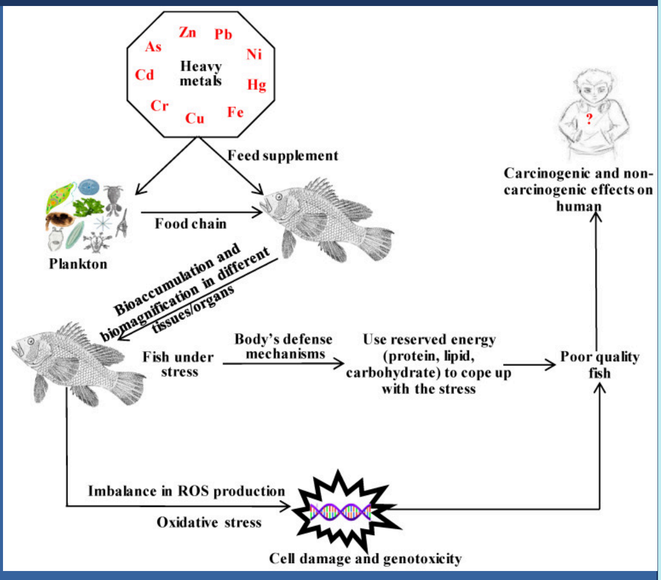


Fig. [1.1]: The process of bioaccumulation and subsequent poisoning of people through fish

NORTH MAKES UP THE MAJORITY OF SURVEY 'REGIONS'

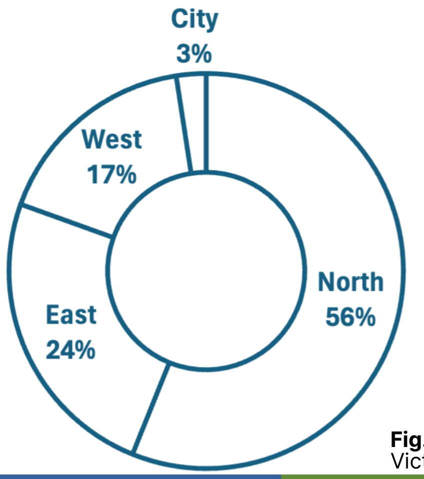
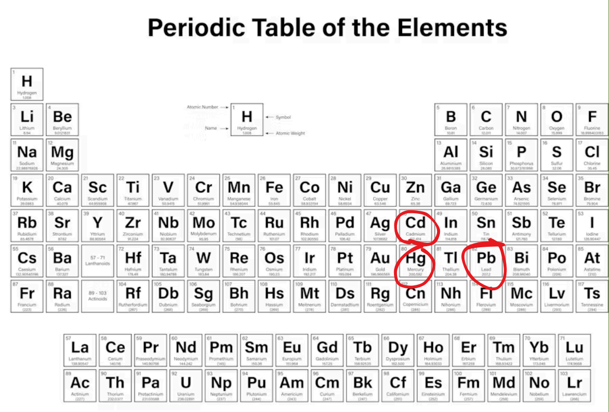


Fig. [1.2]: Population distribution of Victorian fishers

SCOPE OF INVESTIGATION

This investigation examines freshwater health across selected Victorian lakes by measuring factors such as temperature, etc, that affect biodiversity. **[4]**

Primary data focused on lead and mercury, while secondary data from the **EPA** provided insight into cadmium levels. **[3]**



CONTEXT

Investigation into the impact of heavy metals in bodies of fresh water was primarily taken inside the Northern suburbs of Melbourne.

This was done due to the proximity to fish stocking sites and industrial activity as seen in Figure 2.3 and the survey results.

(89% of Northern Suburbs residents recorded living in proximity to Industrial activity.)

ADDENDUM

"Greenvale Reservoir, a key northern water source, showed low biodiversity **despite favourable conditions.**" **[2]**

During experimentation, result showed similar outcomes, most likely due to **heavy chemical treatment.**

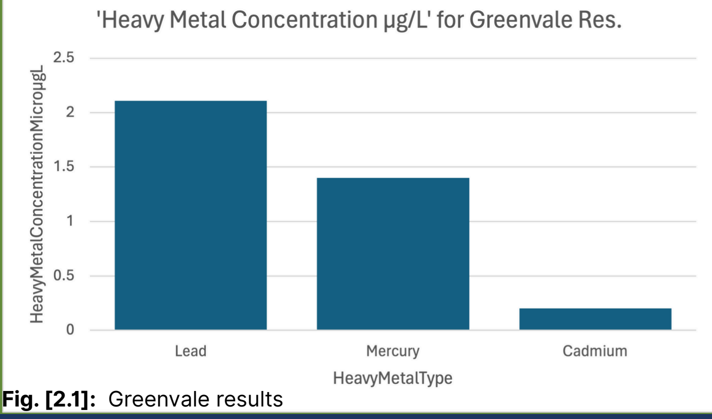


Fig. [2.1]: Greenvale results

FUN FACT

"LD50, or median lethal dose, is a measure in toxicology that indicates the amount of a substance required to kill 50% of a tested population."

If you are in the north, don't drink the water. [3]

DISCUSSION

Surprisingly, results found only a -0.2341 **(Weak)** Pearson correlation between Metal Concentration and Biodiversity.

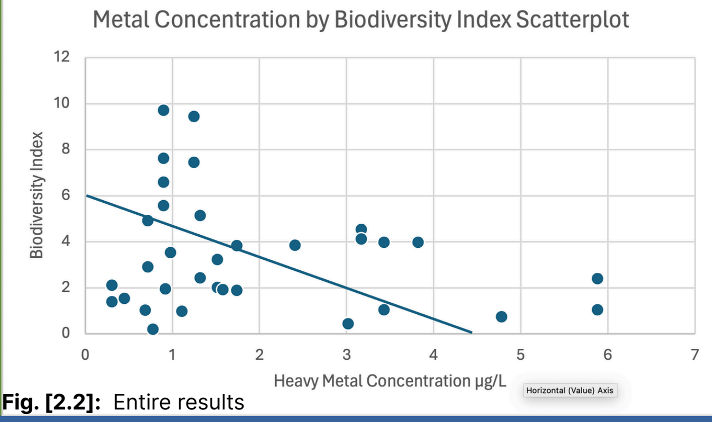
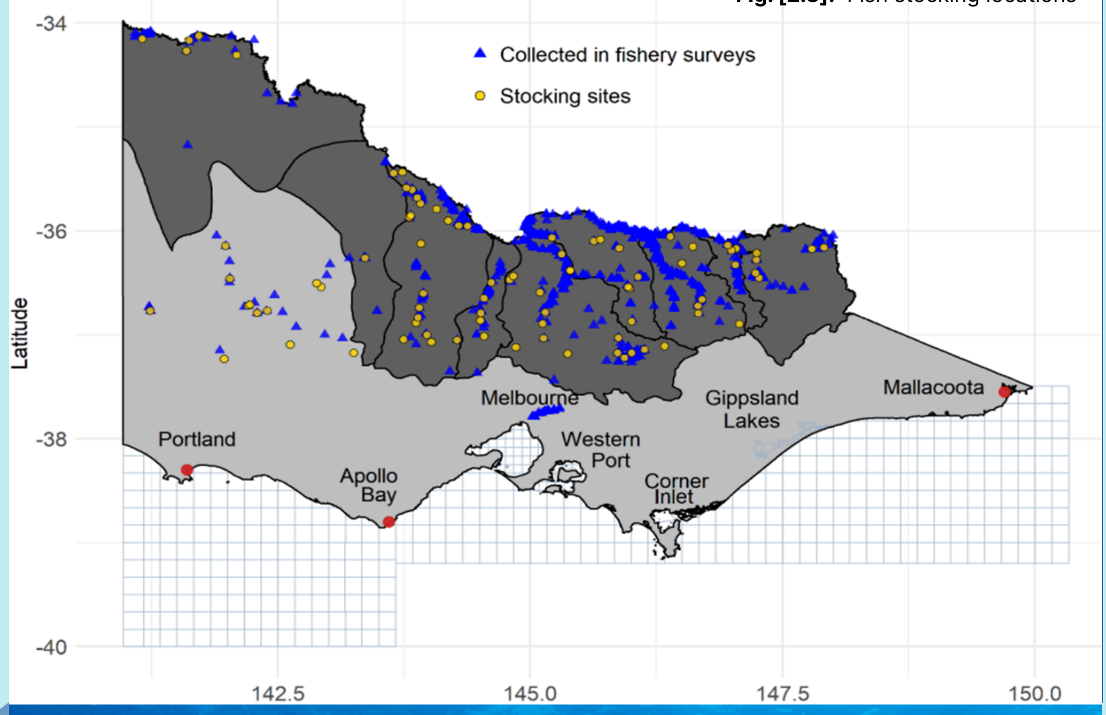


Fig. [2.2]: Entire results

This is in stark opposition to data provided by secondary sources and other journals that discuss the affect heavy metals have on biodiversity. **[5] [7]**

Given short time frames and inexperience in the background of the experimenters that performed the tests in the Northern Suburbs, **experimental errors were bound to arise and affect results**, and the results shown here are likely to be inaccurate.

Fig. [2.3]: Fish stocking locations



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CORRELATION

Results found a -0.7842 Pearson correlation **(Strong)** between Temperature and Biodiversity

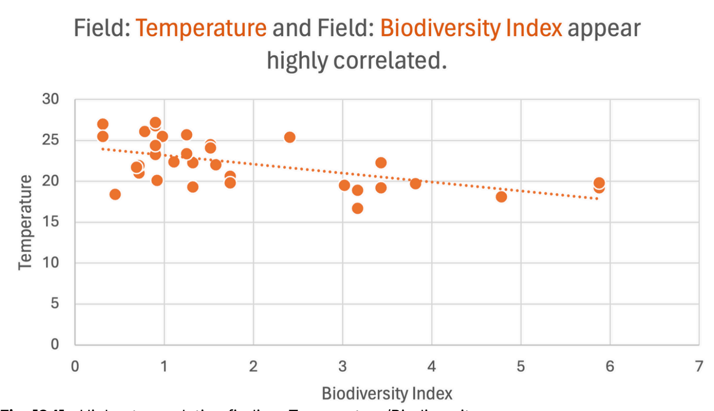

Fig. [3.1]: Highest correlation finding, Temperature/Biodiversity

Further experimentation found a strong correlation between the temperature of freshwater bodies and the biodiversity of species within, aligning well with secondary data and general consensus within surveys.

Strangely enough, compared to other factors such as the turbidity, metal concentration, and the pH level of a water sample, none had any semblance of correlation between the measurement and the biodiversity of a fresh water source. **[6]**

Even when measured against similar occasions/conditions across the globe and in similar climates.
(as seen in Türkiye, Egypt, China, even Aegean sea) [5] [7] [8]

FUN FACT

Of the 4.2 Million recreational fishers in Australia, roughly **72%** of them reside inside Victoria, and, from survey results, approximately **89%** of Victorian fishers live within the **Northern Suburbs.**
[2] [4] [1]

CONCLUSION

This investigation explored the relationship between heavy metal concentrations and fish biodiversity in Victorian freshwater ecosystems, with a focus on the Northern Suburbs of Melbourne.

Contrary to initial expectations, the results revealed only a weak Pearson correlation **(-0.2341)** between metal concentration and biodiversity **(Fig. 2.2)**, challenging secondary data that suggested stronger impacts. **[5][7][8]**

The study identified a strong correlation **(-0.7842)** between temperature and biodiversity, aligning with global trends observed in regions like Türkiye, Egypt, and China. **[5][7]**

Other factors in the primary data however (turbidity, pH) showed no significant links, despite survey data contradicting this 'fact'. **[6]**

Ultimately, these findings highlight the need for further research, especially given potential experimental limitations in human error, skill limitation, etc.

Addressing discrepancies between local and global data however does answer the question.

The presence of heavy metals in bodies of freshwater IS correlated with decreased fish biodiversity.

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